

RCED Model v4.1

Network of Democratic Economic Cells

Analytical Document — Complete and Up-to-Date Version

- The 7 founding building blocks of the model
- New Feature 1: consumer cooperatives that plan
- New Feature 2: 0% interest investment for propagation
- New Feature 3: working hours indexed to automation
 - References, precedents and honest caveats
- What is substantiated, what remains hypothetical

Analytical version: every claim distinguishes documented fact from hypothesis.

Inspired by Piketty, Saez, Zucman, EducationLibre, Mondragon, Migros/Coop, Raiffeisen.

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1. Overview — the model in one page

The RCED model (Network of Democratic Economic Cells) is an architecture for economic transformation that combines a diagnosis of inequality (Piketty, Saez, Zucman), a cooperative productive structure (Social and Solidarity Economy), a decentralised fiscal system inspired by Swiss federalism, and peer-to-peer education (EducationLibre). This v4 adds two elements that complete the architecture: consumer cooperatives that plan their supply, and 0% interest investment that enables the model's propagation.

The logic in one sentence

We make wealth visible (blockchain), cantons tax it and distribute it in three shares, the sovereign fund invests in production AND consumption cooperatives that coordinate democratically, the whole thing governed locally by citizens educated by one another — and once the loop is established, the surplus finances the creation of identical cells elsewhere, at 0% interest.

The 10 layers of the complete model

Layer	Element	Function
0 — Diagnosis	Piketty / Saez / Zucman	Document inequality and tax evasion
1 — Infrastructure	Global blockchain	Makes wealth visible and taxable
2 — Capture	Inverted cantonal taxation	Cantons collect, 40/35/25 key
3 — Capitalisation	Citizens' sovereign fund	Invests in the real economy
4a — Production	Workers' cooperatives	What we make and how
4b — Consumption	Consumer cooperatives	Plan supply (NEW)
5 — Health / Time	CHCI + 4-day week	Cooperative health, inclusion, freed time
6 — Coordination	Democratic coordination	Federated cells, market as signal
7 — Governance	Swiss model	Subsidiarity, communes to Confederation
8 — Knowledge	P2P Education (EducationLibre)	Autonomous citizens, reciprocity credits
9 — Propagation	0% interest investment	Creates cells elsewhere (NEW)
10 — Time	Work / automation indexing	Working hours fall as machines take over (NEW)

2. The 7 founding building blocks (synthetic overview)

These building blocks have been documented in detail in previous reports. Here is a synthetic overview to situate the two new features in context.

Global wealth blockchain

A universal registry making wealth observable, and therefore taxable. A logical extension of Zucman's work on traceability. To be distinguished from a cryptocurrency: this is a fiscal transparency infrastructure, not a speculative asset.

Inverted cantonal taxation (40/35/25 key)

Cantons collect progressive tax and distribute: 40% stays in the canton, 35% goes to the sovereign fund, 25% to fiscal equalisation. An inversion of the usual fiscal chain, consistent with existing Swiss federalism.

Citizens' sovereign fund

Public capitalisation with democratic governance, investing in the real economy rather than speculating. Reference models: Norway, Alaska, Singapore.

Social Economy — workers' cooperatives

Workers' cooperatives, mutual societies, associations with shared governance. This is the production side of the cooperative economy.

CHCI — cooperative health

Cooperative health insurance funds replacing the private component of the LAMal, with premiums proportional to income. Directly connected to cooperative pharmacies (New Feature 1).

Reconfigured work

A 4-day week and active inclusion of unemployed and sick people within cooperative structures. The freed time makes citizen governance and P2P learning possible.

P2P Education (EducationLibre)

A peer-to-peer learning system inspired by BitTorrent, with reciprocity credits NOT convertible into money. Makes citizens capable of understanding and governing the system.

3. New Feature 1 — Consumer cooperatives

Until now, the RCED model dealt mainly with production (who makes things, and how). But one question remained without a clear answer: what signal tells production cooperatives what to produce? The answer is consumer cooperatives.

3.1 The principle: planning by demand, not by supply

Rather than a central state planning production (the historical mistake of the Soviet Gosplan), it is the consumer cooperatives — supermarkets, pharmacies — that plan their supply. They aggregate the real demand of people, as close to the ground as possible. They do not have to guess at needs: they observe them directly in what their members buy.

NEW

The problem of economic calculation is resolved by the right actor. Hayek and von Mises were right: a centre cannot know the dispersed needs of each locality. But consumer cooperatives are in direct contact with need. They hold the local information that the centre could not aggregate.

NEW

The pharmacy case is particularly relevant. Medication is a good in which the pure market often fails: recurring shortages, opaque pricing, dependence on a handful of suppliers. A pharmaceutical cooperative that plans its supply at cantonal scale could pool stocks, smooth shortages, negotiate prices collectively and prioritise according to real health needs. It connects directly to the CHCI (cooperative health).

NEW

Aggregated demand orients production. This is the missing link: the signal that tells production cooperatives what to make comes neither from a state plan, nor solely from the market — it comes from consumer cooperatives that democratically relay real needs from the field.

3.2 Why Switzerland is the ideal terrain

This is not a utopia: Switzerland has already done this at national scale, and these structures dominate its economy. Migros and Coop are consumer cooperatives that effectively plan their supply chains, including by integrating their own production. The RCED model therefore proposes to generalise and democratise an already proven model, extending it to other sectors such as pharmacy.

4. New Feature 2 — 0% interest investment

Once the core RCED model is operational and generating a surplus, how does this model propagate? Not by compulsion, not by predatory debt — by 0% interest investment.

4.1 The principle: propagate, not extract

When the IMF or World Bank lend to a country, they extract interest and impose conditions (austerity, privatisations). The RCED sovereign fund does the opposite: it lends without interest, with the sole aim of reproducing the model elsewhere. This is not naive philanthropy — it is a calculation: each RCED cell that succeeds elsewhere strengthens the legitimacy and resilience of the entire network.

NEW

Three concentric circles of propagation First, neighbouring cantons (cultural proximity, easiest: Jura, Vaud, Berne). Then border regions (France, Italy, Germany). Then the Global South, where the absence of predatory debt is a decisive argument against Chinese or Western competition.

NEW

Each cell in turn becomes an investor This is the viral propagation mechanism. A successful cell does not remain dependent: it generates its own surplus and in turn finances the next cell. This is exactly how Raiffeisen savings banks multiplied in the 19th century, or how Mondragon spread.

NEW

Propagation by example, not by imposition The model being open source (like EducationLibre, under a free licence), any region can adopt and adapt it to its context. Diffusion is organic: no one is forced, we show that it works and finance those who want to try.

4.2 The critical condition: a real and lasting surplus

CAUTION

Essential nuance: 0% interest investment assumes that the sovereign fund generates a real and lasting surplus. As long as the core model has not demonstrated that it produces this surplus, this propagation layer remains purely conditional. We cannot begin by exporting: a first cell must first prove that it works and generates a surplus. The sequence is non-negotiable.

5. New Feature 3 — Working hours indexed to automation

The 4-day week is a sound measure, backed by real data. But a question arises: how far can working hours be reduced? The v4.1 model's answer is not an arbitrary figure (3 days, 2 days) but a principle: working hours fall at exactly the rate at which automation takes over.

5.1 The principle: index, don't decree

The naive version would say: let us decree a 3-day week. The problem: without real productivity gains, reducing working hours causes production to fall, and therefore the tax base, and therefore the sovereign fund's surplus — and the entire propagation layer becomes unaffordable. We would be sawing the branch on which the model sits.

The indexed version inverts this logic: working hours only fall insofar as machines produce in our place. It is the measured consequence of a real gain, not a bet on the future. Robotisation, instead of destroying jobs, frees time — without loss of production or income.

NEW

The productivity gain splits in two. When automation increases output per hour worked, this gain is shared: one part is socialised via the sovereign fund (so that income does not fall when hours fall), and the pace of reduction in working time is decided democratically by the cooperatives themselves — not imposed from above.

NEW

The robot becomes an ally, not a threat. In current capitalism, the machine replaces the worker, the worker is made redundant, and the employer keeps the gain: this is the nightmare version of automation. The model proposes the opposite: the machine does part of the work, and this gain is shared as free time for everyone, without loss of income. The same automation that destroys jobs in capitalism frees time in RCED.

NEW

Work sharing completes the indexing. Rather than simply reducing hours, we redistribute work among more people: the same output achieved by more people each working less. This maintains production while freeing time for everyone — and it directly absorbs the unemployed people inclusion strand of the model.

5.2 The theoretical precedent: Keynes, 1930

DOCUMENTED FACT

Keynes had predicted a 15-hour week — he was wrong about distribution. In 1930, in *Economic Possibilities for our Grandchildren*, John Maynard Keynes imagined that technical progress would allow his grandchildren to work around 15 hours per week. Productivity has indeed exploded since then — Keynes was not wrong on this point. He was wrong about the DISTRIBUTION of its fruits: instead of being converted into free time, the gains were converted into increased consumption and inequality. The RCED model proposes precisely the mechanism that Keynes was missing: a device that captures the productivity gain and redistributes it as time, instead of letting it flow to capital. Source: J. M. Keynes, *Economic Possibilities for our Grandchildren* (1930). Data on the evolution of productivity and working hours since 1930 are widely documented (notably by the OECD and research on the productivity/wages decoupling).

5.3 Conditions for viability

CAUTION

Essential nuances. First, not all sectors automate at the same rate: care, education, and crafts are resistant to automation. The reduction in working hours will therefore be uneven across occupations, and a form of free-time equalisation will be needed (just as

there is fiscal equalisation), on pain of creating a new inequality. Second, automation requires an initial investment in machines, which must come from the sovereign fund and not from private debt — otherwise it is the creditors who capture the gain. Finally, a 3-day week or less remains a conditional horizon: it only becomes possible when productivity genuinely allows it, never before.

6. How it all fits together — nested feedback loops

The v4 model operates through two nested loops: an internal loop (production–consumption) and an external loop (propagation).

6.1 The internal loop: production – consumption

Consumer cooperatives observe real demand and plan their supply. This aggregated demand feeds back to production cooperatives, which adjust what they make. The market remains an information signal (prices indicate scarcities), but it is no longer master: the final decision belongs to the cells, democratically. This can be called democratic productive coordination, rather than centralised planning.

6.2 The external loop: propagation through surplus

Once the internal loop is stabilised, the sovereign fund generates a surplus. This surplus finances, at 0% interest, the creation of RCED cells elsewhere. Each new cell, once mature, generates its own surplus and finances the next. The growing network of cells in turn reinforces the original core (more partners, more legitimacy, more resilience). It is mutual reinforcement, not one-way extraction.

6.3 Why this articulation is coherent

The same principle at two scales

At the local scale, consumer cooperatives orient production through demand. At the continental scale, mature cells orient the creation of new cells through 0% investment. In both cases: information and decision remain as close to the field as possible, and solidarity circulates without extraction.

7. Real-world precedents and references

Each of the two new features rests on documented precedents. Here are the facts substantiated by verifiable sources.

7.1 Consumer cooperatives: Migros, Coop

DOCUMENTED FACT	Consumer cooperatives already dominate Swiss retail Coop was founded in 1864 as a consumer cooperative. Migros was transformed into a cooperative by Gottlieb Duttweiler in 1941, who donated it to its customers. Together they account for a major share of Swiss retail and effectively plan their supply chains, even integrating their own production. The RCED model therefore proposes to democratise and extend what already exists. Sources: Coop Switzerland (history, founded 1864); Migros / Federation of Migros Cooperatives (cooperative transformation 1941, Gottlieb Duttweiler). Public data on market shares in Swiss retail.
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7.2 Cooperative propagation: Raiffeisen, Mondragon

DOCUMENTED FACT	Raiffeisen: propagation by spreading in the 19th century The Raiffeisen savings bank model, founded on cooperation and mutual aid, spread from village to village from the mid-19th century. In Switzerland, Raiffeisen is today one of the leading banking groups, run by its members — proof that the cooperative model can spread and endure.
DOCUMENTED FACT	Mondragon: 70,000 jobs from 5 workers Founded in 1956 by the priest Arizmendiarieta with his students, the Mondragon Corporation employed 70,085 people in 2024 (turnover of €11.2 billion). Of the 103 cooperatives created between 1956 and 1986, only three closed (survival rate of approximately 97% over three decades). Mondragon spread by reinvesting its surpluses — the exact mechanism proposed by 0% investment. Sources: Wikipedia / Mondragon Corporation Annual Report 2024; Participedia; Corporate Rebels (2024); history of the Raiffeisen movement (Friedrich Wilhelm Raiffeisen, 1860s).

8. Honest caveats and limitations

Intellectual integrity requires naming the difficulties of the two new features, not only their promises.

8.1 Limitations of consumer cooperatives

CAVEAT	Upstream dependence does not disappear A consumer cooperative plans its demand, but depends on what producers supply. For medicines, a Swiss cooperative pharmacy remains dependent on the global pharmaceutical industry. Planning demand dampens this dependence, it does not eliminate it — unless the supply chain is climbed (cooperative production of generics).
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CAVEAT	The critical mass risk Migros and Coop are powerful because they are enormous — their negotiating power comes from volume. A small local cooperative has little weight against suppliers. Hence the importance of network logic: autonomous but federated local cooperatives to carry weight together (cooperative purchasing groups).
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CAVEAT	Real rather than formal democratic governance Migros and Coop are today criticised because, having become giants, their cooperative dimension is more formal than lived — few members genuinely participate. The model must avoid this pitfall through P2P education and proximity governance.
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8.2 Limitations of 0% investment

CAVEAT	Everything depends on a surplus that does not yet exist This is the most important limitation: without a real and lasting surplus from the sovereign fund, propagation is impossible. And today this surplus is only a hypothesis — the core model has never been tested at full scale. Propagation is therefore a conditional promise, not an established fact.
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CAVEAT	The risk of paternalism Even at 0% interest, exporting a model to other regions may be perceived as interference or paternalism. The open source character and free (non-imposed) adoption are safeguards, but vigilance remains necessary.
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CAVEAT	Long-term sustainability Lending without interest ties up capital. If too many cells fail or take too long to become autonomous, the source fund may be depleted. The pace of propagation must be cautious and indexed to the real maturation of existing cells.
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8.3 Limitations of working-hours indexing

CAVEAT	Uneven automation creates a risk of new inequality Sectors that automate little (care, education, crafts) will see their working hours fall more slowly. Without free-time equalisation, we create a divide between automatable and non-automatable occupations. This is a real problem the model must address head-on, not a detail.
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CAVEAT	Measuring productivity gains is tricky Indexing working hours to productivity assumes that productivity can be measured — yet productivity is difficult to measure in services and care (what is an extra hour of a teacher's worth?). The mechanism is clear for industry, far more blurred elsewhere.
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9. Implementation sequence

Order matters. Not everything can be done at once. Here is a coherent sequence, from the most immediate to the most distant.

Step	What we build	Condition to proceed	Horizon	
1	Pilot cooperative (production AND consumption) in a commune	Measured results over 18–36 months	Immediate	
2	Cooperative pharmacy linked to an embryonic CHCI	Functional aggregated local demand	Yr 2–4	
3	Democratic coordination among cells in the same canton	Several federated cooperatives	Yr 3–6	
4	Cantonal sovereign fund generating a first surplus	Internal loop stabilised and surplus-generating	Yr 5–10	
5	First 0% investment towards a neighbouring canton	Real and lasting surplus confirmed	Yr 8–12	
6	Network of cells — cross-border propagation	Several autonomous, surplus-generating cells	Yr 12–20	

The guiding principle of the sequence

Each step must prove its value before the next. We do not propagate a model that has not proven itself, we do not lend at 0% a surplus that does not exist, we do not federate cooperatives that do not yet exist. This is exactly the small-steps method that Jean Monnet applied to European construction: concrete institutions that prove their value before extending. The difference: here, the method is democratic from start to finish.

SOURCES — Documented facts: Coop Switzerland (founded 1864); Federation of Migros Cooperatives (1941 transformation, G. Duttweiler); Mondragon Corporation Annual Report 2024; Raiffeisen movement (F. W. Raiffeisen, 1860s); G. Zucman, The Hidden Wealth of Nations (2015) and G20 report (2024); F. Hayek, The Use of Knowledge in Society (1945) on the economic calculation problem. Modelling assumptions: the production–consumption articulation and the 0% propagation mechanism are theoretical constructions of this work, never tested in this combined form. EducationLibre: github.com/nadnone/EducationLibre (CC BY 4.0).